Database Design & Data Modeling

Core Entities:

Entity	Key Fields	Relationships
Teacher	ID, FName, LName, Email (Unique), Password, PhoneNumber, Biography	1 → N with Course, TeacherAssistant
Student	ID, FName, LName, Email (Unique), Password, PhoneNumber	1 → N with Card, StudentCourse
Assistant	ID, FName, LName, Email, Password, PhoneNumber, Biography	N → N with Teacher (via TeacherAssistant)
TeacherAssistant	ID, TeacherID (FK), AssistantID (FK)	Many-to-Many join table
Course	ID, Name, Enrollable (bool), Price (decimal 18,2), TeacherID (FK)	1 → N with Class, Exam, StudentCourse
Class	ID, Name, Price, PDF, HomeworkPDF, CourseID (FK)	1 → N with Video, Quiz
Video	ID, URL, Title, ClassID (FK)	N → 1 with Class
Quiz	ID, Title, ClassID (FK)	1 → N with QuizQuestion
Exam	ID, Title, CourseID (FK)	1 → N with ExamQuestion
Question	ID, QuestionText	1 → N with Choice, ExamQuestion, QuizQuestion
Choice	ID, Text, IsCorrect (bool), QuestionID (FK)	N → 1 with Question
QuizQuestion	ID, QuizID (FK), QuestionID (FK)	Many-to-Many join table (Quiz ↔ Question)
ExamQuestion	ID, ExamID (FK), QuestionID (FK)	Many-to-Many join table (Exam ↔ Question)
StudentCourse	ID, StudentID (FK), CourseID (FK)	Many-to-Many join table (Student ↔ Course) — enrollment
Card	ID, CardholderName, ExpiryDate, CVV, CardNumber, StudentID (FK)	N → 1 with Student (prepared for a future payment phase)
QuizAttempt	ID, StudentID (FK), QuizID (FK), Score, TotalQuestions, SubmittedAt	Records a student's quiz attempt
ExamAttempt	ID, StudentID (FK), ExamID (FK), Score, TotalQuestions, SubmittedAt	Records a student's exam attempt

Key Design Notes:

- Questions are decoupled from Quizzes/Exams through join tables (QuizQuestion, ExamQuestion) — meaning the same question can be reused across multiple assessments.
- The Student ↔ Course relationship (StudentCourse) is a join table representing enrollment, not a direct relationship.
- OnDelete(DeleteBehavior.Restrict) is applied on certain relationships (e.g., StudentCourse→Course and ExamQuestion/QuizQuestion→Question) to avoid the "Multiple Cascade Paths" error in SQL Server.
- Price fields (Course.Price, Class.Price) use decimal precision (18,2) for financial accuracy.
- The Assistant, TeacherAssistant, and Card tables already exist in the schema but are not fully wired into the UI yet (planned for a later phase per TODO.md).

ERD: